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Desigualdad de Young para convoluciones

Proposición 1 (Young). *Sean* $f \in \mathcal{L}^1(\mathbb{R}^n)$, $g \in \mathcal{L}^p(\mathbb{R}^n)$ *con* $p \in [1, \infty]$

$$\implies \|f * g\|_p \leqslant \|f\|_1 \|g\|_p .$$

En particular, $|f * g(x)| < \infty$ *c.t.p.* $x \in \mathbb{R}^n$.

***Demostración*:** Veamoslo en tres casos mutuamente excluyentes:

Caso I: Si $p = \infty$, por la `convolucion/??`, tenemos que $\|f * g\|_\infty \leqslant \|f\|_1 \|g\|_\infty$.

Caso II: Si $p = 1$, por el Teorema de Fubini, tenemos que

$$\begin{aligned}
\|f * g\|_1 &= \int_{\mathbb{R}^n} |f * g(x)| \,\mathrm{d}x = \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y)\, g(y) \,\mathrm{d}y \right| \mathrm{d}x \\
&\leqslant \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)|\, |g(y)| \,\mathrm{d}y \,\mathrm{d}x \overset{\overset{\text{Fubini}}{\downarrow}}{=} \int_{\mathbb{R}^n} |g(y)| \left(\int_{\mathbb{R}^n} |f(x-y)| \,\mathrm{d}x \right) \mathrm{d}y = \\
&\leqslant \int_{\mathbb{R}^n} |g(y)|\, \|f\|_1 \,\mathrm{d}y = \|f\|_1 \|g\|_1 .
\end{aligned}$$

Caso III: Si $p \in (1, \infty)$, tenemos que

$$\|f * g\|_p^p = \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y)\, g(y) \,\mathrm{d}y \right|^p \mathrm{d}x \leqslant \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)| \,\mathrm{d}y \right)^p \mathrm{d}x. \tag{1}$$

Por otro lado, tomando $q \in (1, \infty)$ como el exponente conjugado de p, tenemos que

$$\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)| \,\mathrm{d}y = \int_{\mathbb{R}^n} |f(x-y)|^{1/q}\, |f(x-y)|^{1/p}\, |g(y)| \,\mathrm{d}y.$$

Entonces, por la desigualdad de Hölder para $|f(x-\cdot)|^{1/q}$ y $|f(x-\cdot)|^{1/p}|g|$, tenemos

$$\begin{aligned}
\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)| \,\mathrm{d}y &\leqslant \left(\int_{\mathbb{R}^n} |f(x-y)| \,\mathrm{d}y \right)^{\frac{1}{q}} \left(\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|^p \,\mathrm{d}y \right)^{\frac{1}{p}} = \\
&\leqslant \|f\|_1^{1/q} \left(\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|^p \,\mathrm{d}y \right)^{\frac{1}{p}} .
\end{aligned}$$

Sustituyendo con esta expresión en (**??**), tenemos que

$$\begin{aligned}
\|f * g\|_p^p &\leqslant \int_{\mathbb{R}^n} \left(\|f\|_1^{1/q} \left(\int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|^p \, \mathrm{d}y \right)^{\frac{1}{p}} \right)^p \mathrm{d}x = \\
&\leqslant \|f\|_1^{\frac{p}{q}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|^p \, \mathrm{d}y \, \mathrm{d}x \overset{\substack{\text{Fubini} \\ \downarrow}}{=} \|f\|_1^{\frac{p}{q}} \int_{\mathbb{R}^n} |g(y)|^p \left(\int_{\mathbb{R}^n} |f(x-y)| \, \mathrm{d}x \right) \mathrm{d}y = \\
&= \|f\|_1^{\frac{p}{q}} \|f\|_1 \int_{\mathbb{R}^n} |g(y)|^p \, \mathrm{d}y = \|f\|_1^p \|g\|_p^p .
\end{aligned}$$

Por tanto, elevando a $^1\!/_p$, llegamos a $\|f * g\|_p \leqslant \|f\|_1 \|g\|_p$. ■

Referenciado en

- `lem-transformada-fourier-convolucion`

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